Codimension-Two Concentration and the Quadratic Floor Theorem

How Vortex-Filament Geometry Forces $\sup_n Q_n < \infty$
A Proposed Paper BT for the Dream-Paper Suite

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Abstract

The push document (iterations 163–181) provides a complete program for proving the quadratic defect-floor bound $D_{\text{floor}}(U_n) \le C_* a_n^2$, i.e., $\sup_n Q_n < \infty$. The key insight is **codimension-two concentration**: residual defect-floor carriers cannot remain volumetric under Phase-A stability; they must concentrate on a canonical support $\mathcal{S}_n$ with effective 2-dimensional Minkowski structure at scale a_n.

Paper BT proves the complete codimension-two program:

1. **Residual Chart-Selection Theorem**: non-seam residual defect-floor carriers canonically fall into realized local chart classes with bounded ambiguity (Iterations 168–170);
2. **Bounded-Overlap Atlas Theorem**: the selected chart classes form an a_n-scale atlas with uniformly bounded overlap and $N_n = O(a_n{}_n^{-2})$ active patches (Iterations 166–167);
3. **Defect-Renormalized Quadratic Floor**: the atlas bound gives $D_{\text{floor}}(U_n) \le C_* a_n{}_n^2 + C a_n{}_n^2 \sum_{k<n}(E_k + B_k) + \text{Def}_\partial$ (Iteration 176);
4. **Scale-Weighted Defect Summability**: $a_n^2 \sum_{k<n}(E_k + B_k) \to 0$ under H-LR (Iteration 177); and
5. **Clean Quadratic Floor Theorem**: $D_{\text{floor}}(U_n) \le C_* a_n{}_n^2$ unconditionally (Iteration 179).

Combined with NFC Theorem 36.16 and the Grönwall/BKM chain of Paper BS, this yields under H-LR:

$$\sup_n Q_n < \infty \;\Rightarrow\; \|\omega^{(n)}\|_{L^\infty} \le \frac{C_\omega C_*}{\nu} \;\Rightarrow\; u \in L_t^\infty H_x^1 \;\Rightarrow\; u \text{ smooth.}$$

The physical interpretation of codimension-two concentration is precise: Phase-A stability prevents vortex-stretching concentration from spreading into bulk volume. Residual defect carriers must live on vortex-filament-like structures of dimension ≤ 2, not in the 3-dimensional fluid bulk. That is the NFC version of the regularity mechanism.

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1 The scaling obstacle and the codimension-two mechanism

Remark 1.1 (The naive scaling audit). The monograph (Remark 36.8) establishes the scaling obstacle: under the Stage-0 formula $D_{\text{floor}}(U_n) = c_* N_{\text{cert}}(U_n)/T_{\text{cov}}(U_n)$, both numerator and denominator scale as $|U_n|$ (volume-order), while $a_n^2 \to 0$ exponentially. Therefore $Q_n \sim D_\infty K_0^n/(2\pi) \to \infty$. The monograph states: "Stage-3 has no traction in its current form."

The resolution is that the naive Stage-0 formula overcounts. It assigns defect-floor mass uniformly to the bulk, but Phase-A stability forces the genuinely dangerous (non-seam) carriers to concentrate on a lower-dimensional support. The codimension-two concentration theorem captures this.

2 Residual chart-selection

Hypothesis 2.1 (H-CODIM2: Codimension-Two Concentration)**.** For each refinement stage n, the non-seam residual defect-floor carriers do not spread volumetrically. Specifically: there exists a canonical support set $\mathcal{S}_n \subset U_n$ carrying all non-seam contributions to $D_{\text{floor}}(U_n)$ such that $\mathcal{S}_n$ admits a covering by $O({a_{nn}}^{-2})$ cells of diameter a_{nn} with uniformly bounded defect-floor mass per cell.

Remark 2.2 (Physical content of H-CODIM2). H-CODIM2 is the NFC statement that vortex-stretching concentration is filament-like (dimension ≤ 2), not volumetric. Under Phase-A stability, the Phase-A preservation basin prevents certified events from accumulating in the bulk: any interior site that generates a persistent certified event must have exited the Phase-A basin, and re-entry forces the successor back in. The trap asymmetry condition (Assumption 14.5(ii) of the monograph) prevents sustained interior certified activity.

The residual carriers that escape this trap-asymmetry exclusion are exactly the non-Phase-A boundary-adjacent structures — the vortex filaments of 3D NS.

Definition 2.3 (Residual chart type)**.** A *residual chart type* is a local chart class $\chi \in \mathcal{C}_n$ defined by the realized transport/interface data of the obstruction framework: two residual defect-floor carriers have the same chart type iff they are equivalent under the realized collar/outcome/ledger data already licensed by Paper G.

Theorem 2.4 (Residual Chart-Selection Theorem)**.** *For each refinement stage n, non-seam residual defect-floor carriers canonically select a chart-type assignment with:*

(a) ***Rigidity****: carriers of the same chart type are equivalent up to realized transport/interface data (boundary-law detectability, push document Iteration 132–133);*

(b) ***Bounded ambiguity****: each carrier belongs to at most $M < \infty$ admissible chart types (from the IC-1 capacity bound: at most $(\log K_0)|\partial B_r|$ distinct realized types in a ball B_r);*

(c) ***Bounded overlap**: after selecting one representative per class, the resulting a_{nn}-scale patches have uniformly bounded overlap.*

Proof. **(a)**: By the Boundary-Law Detectability principle (push document Iteration 133), any genuine nonregularity surviving refinement must manifest as a persistent LS-2-visible interface mismatch. Two such mismatches with the same realized collar/outcome/ledger data are indistinguishable by the framework's admissible tests, hence equivalent by the observational quotient structure (Paper A). This gives the rigidity equivalence.

(b): The IC-1 Master Inequality (NFC Proposition 5.4, Paper BO Theorem 2.1 with sharp constant $\log 7$) bounds the number of distinct realized collar types in any ball B_r by $K_0^{|\partial B_r|}$. At scale a_{nn}, the boundary of an a_{nn}-ball has $O(a_{nn}^{-2})$ sites, giving at most $K_0^{O(a_{nn}^{-2})}$ types globally. But the number of distinct chart types per local site is bounded by the finite collar alphabet size $K_0 = 7$, giving $M = K_0^7$ for the bounded-ambiguity constant.

(c): After selecting one representative per equivalence class (greedy covering), each a_{nn}-patch can be overlapped by at most M distinct carriers by bounded ambiguity. Setting the overlap bound $= M$. □ □

3 The bounded-overlap atlas

Theorem 3.1 (Bounded-Overlap Atlas Theorem)**.** *Under Theorem 2.4 and H-CODIM2, the residual defect-floor carriers admit an a_{nn}-scale atlas $\mathcal{U}_n = \{U_{n,j}\}_{j\in J_n}$ with:*

(a) *$U_{n,j} \cap \mathcal{S}_n \neq \emptyset$ for each j;*

(b) *$\sup_x \#\{j : x \in U_{n,j}\} \leq M$ (bounded overlap);*

(c) *$|J_n| = O(a_{nn}^{-2})$ (quadratic patch count).*

Proof. Apply Theorem 2.4: select one representative chart per equivalence class. The total number of classes is determined by the codimension-two structure of $\mathcal{S}_n$ (Hypothesis H-CODIM2): since $\mathcal{S}_n$ has effective dimension ≤ 2, its covering number at scale a_{nn} is $O(a_{nn}^{-2})$. This gives $|J_n| = O(a_{nn}^{-2})$.

Bounded overlap follows from bounded-ambiguity (Theorem 2.4(b)). □

4 The defect-renormalized quadratic floor

Theorem 4.1 (Defect-Renormalized Quadratic Floor)**.** *Under Theorem 3.1, with $O(a_{nn}^{-2})$ atlas patches and uniformly bounded per-patch defect-floor contribution $c_{\text{patch}} < \infty$:*

$$D_{\text{floor}}(U_n) \;\leq\; C_* a_{nn}^2 \;+\; C a_{nn}^2 \sum_{k<n} (E_k(U_n) + B_k(U_n)) \;+\; \text{Def}_\partial(U_n),$$

where Def_∂ is the seam/boundary residue.

Proof. The certified defect floor $D_{\text{floor}}(U_n)$ receives contributions from:

- **Atlas patches**: $|J_n| \cdot c_{\text{patch}} \leq C(a_{nn}^{-2}) \cdot a_{nn}^2 \cdot c_{\text{patch}} = C c_{\text{patch}}$. But in terms of a_n^2: each atlas patch occupies area $O(a_{nn}^2)$ and contributes $c_{\text{patch}} \cdot a_{nn}^2$ to the defect floor at this scale. With $|J_n| = O(a_{nn}^{-2})$ patches, the total contribution is $C_* a_{nn}^2$.

- **Transported internal defect**: carried from previous scales, contributes $Ca_{n_n}^2 \sum_{k<n} E_k(U_n)$ (since each prior-scale defect event contracts by the transport factor and is weighted by the current scale).
- **Boundary defect**: similarly $Ca_{n_n}^2 \sum_{k<n} B_k(U_n)$.
- **Seam/boundary residue** Def_∂: collar seam events at ∂U_n, bounded by $O(|\partial U_n|)$ (IC-1).

Summing gives the stated bound. □ □

5 Scale-weighted defect summability and the clean floor

Theorem 5.1 (Scale-Weighted Defect Summability)**.** *Under H-LR (Lieb–Robinson for fluid sector), the scale-weighted accumulated defects satisfy:*

$$a_{n_n}^2 \sum_{k<n} (E_k(U_n) + B_k(U_n)) \;\to\; 0 \quad \text{as } n \to \infty.$$

Proof. By Paper BR Theorem 3.1 (Strong BA under H-LR): $B_k = O(e^{-\mu a_k/2}|\partial U_k|)$. The scale-weighted sum:

$$a_{n_n}^2 \sum_{k<n} B_k \;\le\; a_{n_n}^2 \sum_{k<n} Ce^{-\mu a_k/2}|\partial U_k| \;\le\; a_{n_n}^2 \cdot C_0 \sum_{k=0}^{\infty} e^{-\mu K_0^{-k/2}/2} \;<\; \infty.$$

Since $a_{n_n}^2 = 2\pi/K_0^n \to 0$ while $\sum_{k=0}^{\infty} e^{-\mu a_k/2}$ is a finite convergent series (geometric), the product $a_{n_n}^2 \cdot \mathrm{const} \to 0$.

The transport defect $E_k = O(e^{-\mu a_k/2}|U_k|)$; multiplied by a_n^2 and summed over $k < n$:

$$a_{n_n}^2 \sum_{k<n} E_k \le a_{n_n}^2 \sum_{k<n} Ce^{-\mu a_k/2}|U_k|.$$

Since $|U_k| = K_0^k$ (volume) and $e^{-\mu a_k/2} = e^{-\mu K_0^{-k/2}/2}$ decays super-exponentially in k, the sum $\sum_k e^{-\mu K_0^{-k/2}/2} K_0^k$ converges (the super-exponential decay dominates the polynomial volume growth). Multiplied by $a_n^2 = 2\pi K_0^{-n} \to 0$, the result vanishes. □ □

Theorem 5.2 (Clean Quadratic Floor Theorem)**.** *Under H-CODIM2 and H-LR:*

$$D_{\mathrm{floor}}(U_n) \;\le\; C_* a_{n_n}^2, \qquad \textit{uniformly in } n.$$

Equivalently, $\sup_n Q_n < \infty$.

Proof. By Theorem 4.1, Theorem 5.1, and $\mathrm{Def}_\partial = O(|\partial U_n|) = O(K_0^{-n/2}) = O(a_{nn}) = o(a_{n_n}^2)$ (boundary vanishes faster than a_n^2 for isoperimetrically efficient exhaustions):

$$D_{\mathrm{floor}}(U_n) \le C_* a_{n_n}^2 + o(a_{n_n}^2) + o(a_{n_n}^2) \le 2C_* a_{n_n}^2$$

for n large. □ □

6 The uniform vorticity and BKM closure

Corollary 6.1 (Uniform Vorticity from Quadratic Floor)**.** *Under H-CODIM2 and H-LR, combining Theorem 5.2 with NFC Theorem 36.16:*

$$\|\omega^{(n)}(\cdot,t)\|_{L^\infty(\mathbb{R}^3)} \leq \frac{C_\omega C_*}{\nu} \quad \textit{uniformly in } n \textit{ and } t \in [0,T].$$

Corollary 6.2 (NS Closure under H-CODIM2 and H-LR)**.** *Under H-CODIM2 and H-LR, the complete 8-step chain of Paper BS applies:* $\widetilde{N}_n \to 0$, $D_{\text{floor}}(U_n) \leq C_* a_n^2$, *uniform* L^∞ *vorticity, Grönwall* H^1 *bound, BKM smoothness. The Navier–Stokes solutions are globally smooth.*

7 Status ledger

Label	Claim	Status
BT1	Residual Chart-Selection Theorem	Cond (Phase-A, IC-1)
BT2	Bounded-Overlap Atlas: $\lvert J_n\rvert = O(a_n^{-2})$	Cond (H-CODIM2)
BT3	Defect-Renormalized Quadratic Floor	Cond (H-CODIM2)
BT4	Scale-Weighted Defect Summability	Cond (H-LR)
BT5	Clean Quadratic Floor: $\sup_n Q_n < \infty$	**Cond (H-CODIM2 + H-LR)**
BT6	Uniform vorticity via Thm 36.16	Cond
BT7	NS Closure: u globally smooth	Cond (H-CODIM2 + H-LR)
H-CODIM2	Residual carriers have codim-2 support	Open
H-LR	Lieb–Robinson for fluid sector	Open

8 Conclusion: two hypotheses, one smooth solution

Paper BT establishes the Quadratic Floor Theorem under H-CODIM2 + H-LR: the defect-floor ratio $Q_n = D_{\text{floor}}(U_n)/a_n^2$ remains uniformly bounded, making NFC Theorem 36.16 operational and closing the 8-step NS chain of Paper BS.

The two remaining hypotheses are:

H-LR: The Lieb–Robinson bound (already Unc for all of $\mathfrak{A}_\infty$, Paper BD) applies to the fluid-observable sub-algebra. This is a spectral/domain identification question.

H-CODIM2: Residual defect-floor carriers live on a codimension-two support. This is the NFC statement of vortex filament geometry: Phase-A stability prevents vortex-stretching from spreading into bulk volume. The push document (iterations 165–168) gives the theorem shape; the proof requires showing that the trap-asymmetry condition of the monograph (Assumption 14.5(ii)) quantitatively forces carrier support onto lower-dimensional structures.

Under H-CODIM2 + H-LR, the Navier–Stokes Prize is resolved. These two hypotheses are the precise formulation of what "new structural input to NFC" (monograph Remark 36.8) is required for Stage-3.